

5(a)	1 prevents formation of, cross links / cross linkages (between, peptidoglycan / murein, chains) ; A peptide cross links A links between, murein / polymer, chains I peptide bonds I formation of peptidoglycan R if cellulose chains stated 2 (penicillin) inhibits, transpeptidase action / enzyme involved in forming cross links ; A alternative correct names for transpeptidase 3 weakens <u>cell wall</u> ; A cell wall unable to withstand (turgor) pressure A cell wall loses strength R idea that penicillin, punches / makes, holes, to weaken 4 (cell), lysis / bursts / ruptures / AW (so bacterium killed) ; 5 acts, on growing bacteria / when bacteria are increasing in size (when cell wall needs to be synthesised) ; I growing, wall / peptidoglycan chains	3
5(b)	(<i>Plasmodium</i> / <i>P.</i>), <u>ovale</u> / <u>falciparum</u> / <u>malariae</u> / <u>vivax</u> ; correct spelling I if <i>Plasmodium</i> is written after the species name if more than one given, all must be correct	1
5(c)	<i>any one valid suggestion:</i> male does not, need protein for <u>egg</u> production / produce <u>eggs</u> ; R larvae I male does not reproduce male does not have mouthparts for piercing skin ; AW e.g. no 'needle' to pierce skin (to suck blood) adult male does not feed ; adult male feeds (only) on, plants / nectar ; blood is toxic to males ; can't detect presence of, humans / mammals ; male does not produce anticoagulant (for blood) ;	1

Question	Answer	Marks
5(d)	accept mosquito or vector for <u>Anopheles</u> accept, pathogen / parasite, for <u>Plasmodium</u> max 2 1 idea that individuals / people, taking antibiotics for bacterial diseases will pass on antibiotics to <i>Anopheles</i> when it feeds ; e.g. blood taken by <i>Anopheles</i> contains antibiotics 2 (so) antibiotics kill bacteria (in <i>Anopheles</i> gut) ; <i>must be in context of gut bacteria</i> 3 decreased / no, competition between, <i>Plasmodium</i> and (gut) bacteria (so more <i>Plasmodium</i> survives) ; 4 higher survival of <i>Plasmodium</i> makes effective (<i>Anopheles</i>) immune response more difficult ; AW (so <i>Anopheles</i> more likely to pass on <i>Plasmodium</i>) max 2 I ref. to antibiotic resistance 5 use of antibiotics may increase, incidence / number of cases of, malaria ; 6 and 7 two marks for examples of what doctors need to consider ; e.g. need to balance antibiotic intake with increased risk of malaria transmission idea that do not want to stop people taking antibiotics / antibiotics needed to fight (bacterial) infections treat for malaria before giving antibiotics for (non-serious / non-life threatening) bacterial infections only prescribe antibiotics that have, no / low, impact on bacteria in <i>Anopheles</i> (gut) (consider) avoiding use of antibiotics to treat malaria 8 AVP ; e.g. need to research which antibiotics have this effect look for alternatives to antibiotics to treat bacterial infections	3

Question	Answer	Marks
5(e)	<i>any four from:</i> 1 (<i>Plasmodium</i>) is a, eukaryote / protist, so has many antigens ; R bacterium / virus, is a eukaryote or (<i>Plasmodium</i>) has many genes coding for (different) antigens ; I antigenic variation 2 <i>idea that</i> different <i>Plasmodium</i> species have different antigens ; I antigenic variation I strains <i>for species</i> 3 (<i>Plasmodium</i>) has different stages of life cycle (within human) with different antigens / shows antigenic variation ; 4 antigenic concealment / <i>Plasmodium</i> spends part of life cycle within host cells / AW ; A short time in blood plasma A spends time inside, red blood cells / liver cells 5 need to find the antigens that give the strongest immune response ; 6 need to, develop / use, more than one type of vaccine ; A cannot use only one type of vaccine 7 AVP ; e.g. mutations will give changed antigens need to find antigens present in, all / most, stages of life cycle <i>difficulties in producing a generic vaccine max 2</i> 8 costly to produce / need to keep costs low / developing countries need to be able to afford vaccine ; 9 needs to have a long shelf life / be stable / be easily stored (e.g. without cold storage) / AW ; 10 (immunity) needs to be long-lasting / aim to avoid boosters / need to develop a single dose vaccine ; AW	4